

Xiangde Luo (罗祥德), PhD

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Bio: Xiangde is a postdoctoral researcher in **Prof. Ruijiang Li's** lab at **Stanford Medicine**. His research lies at the intersection of **computational pathology**, **multimodal foundation models**, and **AI for Science**. He is currently focused on developing multimodal pathology foundation models that integrate histology with spatial transcriptomics and proteomics to characterize how the spatial architecture of the TME shapes disease progression, therapeutic resistance, and clinical outcomes. His long-term goal is to translate these insights into scalable imaging biomarkers for precision oncology. Previously, he developed deep learning methods for annotation-efficient learning and biomedical image analysis. He has also led several community efforts in MICCAI, including the semi-supervised medical image segmentation benchmark **SSL4MIS** (2,600+ GitHub stars), the **SegRap2023** challenge, and two abdominal organ segmentation datasets **WORD** and **RAOS**. His work has received over **16,000 citations**, including **eight papers with ≥ 100 citations**, with an ***h*-index of 30** and an ***i10*-index of 42**. He serves as an **Area Chair** for **MICCAI 2025/2026** and was named to the **Stanford/Elsevier World's Top 2% Scientists 2025**. *If you are interested in his work, have any questions, or would like to explore potential collaboration or work opportunities, please feel free to contact him.*

Education

- 2018-09 – 2024-06  **Ph.D., University of Electronic Science and Technology of China (UESTC)**
Supervisor: **Prof. Shaoting Zhang**
Thesis: *Medical Image Segmentation Using a Limited Amount of Labeled Data*
- 2014-09 – 2018-06  **B.E., University of Electronic Science and Technology of China (UESTC)**
Supervisor: **Prof. Shaoting Zhang**
Thesis: *Data capture using crawl and its applications*

Employment History

- 2024-11 – present  **Postdoctoral Fellow at Stanford University, Advisor: Prof. Ruijiang Li**
- 2024-07 – 2024-10  **Research Assistant at West China Hospital, Mentor: Prof. Kang Li**
- 2021-06 – 2024-06  **Research Intern at Shanghai AI Laboratory, Mentor: Prof. Xiaofan Zhang**
- 2019-07 – 2020-06  **Research Intern at SenseTime Research, Mentor: Dr. Tao Song**

Selected Publications

Journal Articles

- 1 **X. Luo**, J. Xiang, Y. Ji, and R. Li, “nnMIL: A generalizable multiple instance learning framework for computational pathology,” *Nature Biomedical Engineering (In Press)*, 2026.
- 2 H. Wang, ..., and **X. Luo** , “Versatile source-free active domain adaptation for multi-center and multi-rater medical image segmentation,” *Information Fusion*, p. 103 586, 2025.
- 3 **X. Luo**, J. Fu, Y. Zhong, *et al.*, “SegRap2023: A benchmark of organs-at-risk and gross tumor volume segmentation for radiotherapy planning of nasopharyngeal carcinoma,” *Medical Image Analysis*, vol. 101, p. 103 447, 2025.
- 4 **X. Luo**, H. Wang, J. Xu, *et al.*, “Generalizable magnetic resonance imaging-based nasopharyngeal carcinoma delineation: Bridging gaps across multiple centers and raters with active learning,” *International Journal of Radiation Oncology* Biology* Physics*, vol. 121, no. 5, pp. 1384–1393, 2025.
- 5 **X. Luo**, X. Wang, F. Eweje, *et al.*, “Ensemble learning of foundation models for precision oncology,” *arXiv preprint arXiv:2508.16085*, 2025.

- 6 H. Wang, ..., and **X. Luo** , "Dual-reference source-free active domain adaptation for nasopharyngeal carcinoma tumor segmentation across multiple hospitals," *IEEE Transactions on Medical Imaging*, vol. 43, no. 12, pp. 4078–4090, 2024.
- 7 **X. Luo**, W. Liao, Y. Zhao, *et al.*, "A multicenter dataset for lymph node clinical target volume delineation of nasopharyngeal carcinoma," *Nature Scientific Data*, vol. 11, p. 1085, 2024.
- 8 **X. Luo**, W. Liao, Y. He, *et al.*, "Deep learning-based accurate delineation of primary gross tumor volume of nasopharyngeal carcinoma on heterogeneous magnetic resonance imaging: A large-scale and multi-center study," *Radiotherapy and Oncology*, vol. 180, p. 109 480, 2023.
- 9 **X. Luo**, W. Liao, J. Xiao, *et al.*, "WORD: A large scale dataset, benchmark and clinical applicable study for abdominal organ segmentation from ct image," *Medical Image Analysis*, vol. 82, p. 102 642, 2022.
- 10 **X. Luo**, T. Song, G. Wang, *et al.*, "SCPM-Net: An anchor-free 3d lung nodule detection network using sphere representation and center points matching," *Medical Image Analysis*, vol. 75, p. 102 287, 2022.
- 11 **X. Luo**, G. Wang, W. Liao, *et al.*, "Semi-supervised medical image segmentation via uncertainty rectified pyramid consistency," *Medical Image Analysis*, vol. 80, p. 102 517, 2022.
- 12 **X. Luo**, G. Wang, T. Song, *et al.*, "MIDeepSeg: minimally interactive segmentation of unseen objects from medical images using deep learning," *Medical Image Analysis*, vol. 72, p. 102 102, 2021.

Conference Proceedings

- 1 Z. Luo, ..., and **X. Luo** , "Dynamic gradient sparsification training for few-shot fine-tuning of ct lymph node segmentation foundation model," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2025.
- 2 **X. Luo**, Z. Li, S. Zhang, W. Liao, and G. Wang, "Rethinking abdominal organ segmentation (raos) in the clinical scenario: A robustness evaluation benchmark with challenging cases," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2024, pp. 531–541.
- 3 **X. Luo**, M. Hu, W. Liao, *et al.*, "Scribble-supervised medical image segmentation via dual-branch network and dynamically mixed pseudo labels supervision," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2022, pp. 528–538.
- 4 **X. Luo**, M. Hu, T. Song, G. Wang, and S. Zhang, "Semi-supervised medical image segmentation via cross teaching between cnn and transformer," in *International Conference on Medical Imaging with Deep Learning*, PMLR, 2022, pp. 820–833.
- 5 **X. Luo**, J. Chen, T. Song, and G. Wang, "Semi-supervised medical image segmentation through dual-task consistency," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 35, 2021, pp. 8801–8809.
- 6 **X. Luo**, W. Liao, J. Chen, *et al.*, "Efficient semi-supervised gross target volume of nasopharyngeal carcinoma segmentation via uncertainty rectified pyramid consistency," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2021, pp. 318–329.

Research topics

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|--------------------|---|
| AI for Science |  Multimodal pathology foundation model, spatial transcriptomics and proteomics |
| Precision Oncology |  Imaging biomarker discovery, prognosis, immunotherapy response prediction |

Academic Experience and Achievements

Awards and Achievements

- 2023  TMI Distinguished Reviewer
- 2022  MICCAI Student Travel Award

Journals/Conferences Review

- Journals  Science, Nature, Lancet Digital Health, Nature Communication, TPAMI, MedIA, TMI, TNNLS, TGRS, PR, JBHI, eClinicalMedicine
- Area Chair  MICCAI2025, MICCAI2026
- Conferences  MICCAI2021-26, AAAI2022-2025, CVPR2023-2025, ICCV2023/2025